

Brad Berryman
Site Vice President

Susquehanna Nuclear, LLC
769 Salem Boulevard
Berwick, PA 18603
Tel. 570.542.2904 Fax 570.542.1504
Brad.Berryman@TalenEnergy.com



AUG 18 2017

U. S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

10 CFR 50.73

SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-388(387)/2017-005-00
UNIT 1 LICENSE NO. NPF-14
UNIT 2 LICENSE NO. NPF-22
PLA-7618

Docket No. 50-387
50-388

Attached is Licensee Event Report (LER) 50-388(387)/2017-005-00. The LER reports an event involving the loss of Secondary Containment due to a trip of Zone III Filtered Exhaust fan. This event was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(v)(C) as a condition that could have prevented fulfillment of a safety function.

Should you have any questions regarding this submittal, please contact Mr. Jason Jennings, Manager – Nuclear Regulatory Affairs at (570) 542-3155.

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new regulatory commitments.

A handwritten signature in blue ink, appearing to read "B. Berryman", with a long horizontal line extending to the right.

B. Berryman

Attachment: LER 50-388(387)/2017-005-00

Copy: NRC Region I
Ms. T. E. Hood, NRC Project Manager
Ms. L. H. Micewski, NRC Sr. Resident Inspector
Mr. M. Shields, PA DEP/BRP

NRC FORM 366 (04-2017)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104		EXPIRES: 03/31/2020			
 LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NRC-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.										
1. FACILITY NAME Susquehanna Steam Electric Station Unit 2					2. DOCKET NUMBER 05000388			3. PAGE 1 OF 3		
4. TITLE Secondary Containment Declared Inoperable Due to Trip of Zone III Filtered Exhaust Fan										
5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME Susquehanna Steam Electric Station Unit 1	DOCKET NUMBER 05000387
06	24	2017	2017	- 005	- 00	06	18	2017	FACILITY NAME	DOCKET NUMBER 05000
9. OPERATING MODE 1		11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)								
		☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		
		☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		
		☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		
		☐ 20.2203(a)(2)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		
10. POWER LEVEL 100		☐ 20.2203(a)(2)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)		
		☐ 20.2203(a)(2)(iii)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)		
		☐ 20.2203(a)(2)(iv)		☐ 50.46(a)(3)(ii)		☒ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)		
		☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(i)		
		☐ 20.2203(a)(2)(vi)		☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(ii)		
				☐ 50.73(a)(2)(i)(C)		☐ OTHER		Specify in Abstract below or in NRC Form 366A		
12. LICENSEE CONTACT FOR THIS LER										
LICENSEE CONTACT R. B. Genovese - Nuclear Regulatory Affairs								TELEPHONE NUMBER (Include Area Code) (570) 542-2980		
13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT										
CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	
B	NG	FSV	A499	Y						
14. SUPPLEMENTAL REPORT EXPECTED					15. EXPECTED SUBMISSION DATE			MONTH	DAY	YEAR
☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO										
ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)										
On June 24, 2017, at approximately 1028 hours, Unit 2 secondary containment Zone III differential pressure was not maintained above the Technical Specification (TS) limit of 0.25 in w.g. due to the trip of a Reactor Building (RB) Zone III Filtered Exhaust Fan. As a result, TS 3.6.4.1 Surveillance Requirement (SR) 3.6.4.1.1 was not met due to the loss of differential pressure. The condition is being reported in accordance with 10 CFR 50.73(a)(2)(v)(C) as an event or condition that could have prevented fulfillment of a safety function. The cause of the event was the failure of the ASCO solenoid valve. Based on further review, the ASCO solenoid valve has been determined to be less than adequate for its application in Reactor Building HVAC. There were no actual consequences to the health and safety of the public as a result of this event.										

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Susquehanna Steam Electric Station Unit 2	05000388	2017	- 005	- 00

NARRATIVE**CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 100 percent Rated Thermal Power

Unit 2 – Mode 1, approximately 100 percent Rated Thermal Power

There were no structures, systems, or components that were inoperable at the start of the event that contributed to the event.

EVENT DESCRIPTION

On June 24, 2017, at approximately 1028 hours, Unit 2 Reactor Building [EIS System Code: NG] secondary containment Zone III differential pressure was not maintained above the Technical Specification (TS) limit of 0.25 in w.g. A Nuclear Plant Operator (NPO) reported Filtered Exhaust System Fan A [EIS Code: FAN] had tripped. The NPO then placed Filtered Exhaust System Fan B [EIS Code: Fan] in START and Filtered Exhaust System Fan A in STANDBY. Zone III differential pressure was restored at approximately 1042 hours.

On June 24, 2017, at 1542 hours, this condition was reported (ENS #52826) in accordance with 10 CFR 50.72(b)(3)(v)(C) as an event or condition that, at the time of discovery, could have prevented the fulfillment of the safety function. SSES has no redundant Secondary Containment System.

The cause of the trip was due to a failed solenoid valve [EIS Code: SOL] on the discharge damper [EIS Code: DMP]. Upon failure of the solenoid valve, the discharge damper failed closed and Filtered Exhaust System Fan A tripped on low flow.

CAUSE OF EVENT

The cause of the event was the failure of the ASCO solenoid valve. Based on further review, the ASCO solenoid valve has been determined to be less than adequate for its application in Reactor Building HVAC.

ANALYSIS/SAFETY SIGNIFICANCE

Technical Specification 3.6.4.1 Surveillance Requirement 3.6.4.1.1 requires that secondary containment vacuum is maintained greater than or equal to 0.25 in. w.g. This event is being reported pursuant to 10 CFR 50.73(a)(2)(v)(C), as a loss of Secondary Containment differential pressure results in a condition that could have prevented fulfillment of a safety function to mitigate the consequences of an accident by controlling the release of radioactive material.

Based on engineering analysis of the event, secondary containment could have performed its safety function of isolating as assumed in the accident analysis and of re-establishing 0.25 in w.g. vacuum (drawdown) within the assumed accident analysis time (10 minutes).

This event will not be counted as a safety system functional failure (SSFF) for the NRC performance indicator based on the engineering analysis supporting the system's ability to fulfill the safety function.

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CONTINUATION SHEET**

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Susquehanna Steam Electric Station Unit 2	05000388	YEAR 2017	SEQUENTIAL NUMBER - 005	REV NO. - 00

CORRECTIVE ACTIONS

Key corrective actions include replacement of the failed solenoid valve with a more robust model.

COMPONENT FAILURE INFORMATION

Manufacturer: ASCO Services
Type: 3-way electrical solenoid valve
Process Fluid: Air
Model: ASCO 8320 Solenoid Valve, General Use Model 8320G003
Voltage: 120 VAC

PREVIOUS SIMILAR EVENTS

The following are recent LERs involving loss of secondary containment due to component failures:

LER 50-387(388)/2016-012-00, "Unit 2 HVAC Unable to Maintain Differential Pressure," dated May 26, 2016.

LER 50-387(388)/2016-003-00, "Unit 2 Zone 3 HVAC Unable to Maintain Differential Pressure," dated May 6, 2016.

LER 50-387(388)/2015-012-00, "Loss of Secondary Containment Differential Pressure Due to Icing of the Intake Supply Plenum Screens," dated February 2, 2016.

LER 50-387(388)/2015-013-00, "Loss of Secondary Containment Due to Failure of Fans," dated February 2, 2016.

LER 50-387(388)/2015-010-00, "Loss of Zone 2 During Unit 1 Reactor SCRAM," dated January 8, 2016.

LER 50-387(388)/2015-005-00, "Loss of Secondary Containment Due to Unit 2 Damper Alignment," dated June 25, 2015.

LER 50-387(388)/2015-002-00, "Loss of Secondary Containment due to failure of running fans," dated June 10, 2015.